

Tide retrospective: susceptibilities recover the anharmonic coefficients

Tide loop (Claude Fable 5, with GPT-5.6 Sol deliberation)

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1 Setting

Every recovery tide so far certified Section 7.4 of the germbij note (the singular, constructive side). The note’s *regular*-case statement, Theorem 3.1, says the expansions over all observables determine the Taylor coefficients of the loss at a nondegenerate minimum by a triangular induction pairing coefficients against susceptibilities. The seabed’s Stage-2 anharmonic programme (from the primer work in May) already carries the two limits that drive the induction’s first two rungs; this tide reads them backwards.

2 The thing formalised

One theorem in `Laplace/OneD/RecoveryAnharmonic.lean`¹, sorry-free with zero warnings. For the anharmonic potential $\frac{\lambda}{2}x^2 + \frac{\alpha}{6}x^3 + \frac{\gamma}{24}x^4$ (positive λ, γ , discriminant $\alpha^2 < 3\lambda\gamma$), the seabed knows that the mean susceptibility $\partial_h \langle x \rangle_h|_{h=0}$ tends to $-1/\lambda$ and the rescaled covariance susceptibility to α/λ^3 . The theorem: if these two observable data streams agree eventually for two parameter triples, then $\lambda_1 = \lambda_2$ and $\alpha_1 = \alpha_2$. This is the note’s Theorem 3.1 recovery, certified at orders two and three: the data of the expansions pins the jet, coefficient by coefficient. The quartic coefficient γ is the next rung and needs the fourth-cumulant asymptotic, which the seabed does not yet carry.

3 Proof strategy

Entirely the uniqueness-of-limits pattern the thread has now used three times: transport each limit along the eventual equality (`Tendsto.congr'`), identify the limits (`tendsto_nhds_unique`), and extract the parameters by field arithmetic ($-1/\lambda$ is injective on positives; then α/λ^3 with λ pinned). The susceptibility data are treated as opaque functions of t throughout; nothing about the underlying Gibbs derivatives is unfolded.

4 Roadblocks and resolutions

None: the first first-attempt-clean build of the recovery thread (and the second of the germbij programme). The consult confirmed the statement packaging (two separate eventual-equality hypotheses rather than a pair) before the build.

¹`anharmonic_susceptibility_recovery`.

5 What was Mathlib, what was new

Mathlib: uniqueness of limits and filter congruence. Seabed: the Stage-2 anharmonic asymptotics, consumed for the first time by the recovery thread; the discriminant hypothesis and the Threepoint-to-Laplace bridges arrive with them. New: only the reading direction.

6 Lessons

When a thread's pattern stabilizes (this is the third uniqueness-of-limits recovery), the marginal tide is nearly free: the whole cost was locating the right seabed limits. The productive act was the survey, not the proof.

7 Follow-ups

- The γ rung: a fourth-cumulant asymptotic ($t^3\kappa_4 \rightarrow \text{explicit value}$) in the Stage-2 style, which would extend both the primer programme and this recovery.
- Tag the note's Theorem 3.1 with the recovery theorem once merged (the note's regular case now has its first certified instance).